

< T E S T >

Nov. 3th, 2021

150 minutes / Closed Book

Calculator & 2-Page Summary Sheet Allowed

Problem 1 [10pt, 5pt each]

Two tributaries 1 and 2 merge to a main channel 3. Assume that tributaries 1 and 2 have equal bankful capacities, however, hydrological characteristics of the two corresponding drainage basins are somewhat different. During storm events, $P(E_1) = P(\text{Bankful capacity of tributary 1 is exceeded}) = 0.10$, $P(E_2) = P(\text{Bankful capacity of tributary 2 is exceeded}) = 0.20$. Also we know that $P(E_1|E_2) = 0.50$ and $P(E_2|E_1) = 1.00$.

- If the bankful capacity of the main channel 3 is the same as tributaries 1 and 2, what is the probability that bankful capacity of the main channel is exceeded? (Assume that when tributaries 1 and 2 are carrying less than their bankful capacities, flow in main channel 3 exceeding its bankful capacity is 0.2).
- If the bankful capacity of the main channel 3 is twice that of tributaries 1 and 2, what is the probability that bankful capacity of the main channel is exceeded? (Assume that, if only tributaries 1 or 2 exceeds its bankful capacity, flow in main channel 3 exceeding its bankful capacity is 0.15).

Problem 2 [10pt, 5pt each]

The impact of pressure sea waves on coastal structures may be evaluated as $Z = cX^2$ where X is horizontal velocity of the advancing wave and c is a constant. Because of the uncertainty involved in the evaluation of X , we consider this to be a random variable; Z is the derived variable from X . Let $Y = (X - \mu_X) / \sigma_X$, which has zero mean, unit variance, and pdf

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} \exp(-0.5y^2)$$

where μ_X and σ_X are the mean and variance of X .

- Find the mean and variance of $W = Y^2$ using the moment generating function approach.
- Express the mean and variance of Z with the mean and variance of X .

Problem 3 [15pt, 5pt each]

- 'The recurrence time' (Tr) in a Bernoulli sequence has a geometric distribution. Explain why.
- The mean recurrence time (= the return period = μ_{Tr}), is equal to the reciprocal of the probability (p) of the event within one time unit, i.e. $\mu_{Tr} = 1/p$, Prove it mathematically.

- (c) An offshore structure is designed for a height of 8 meters above the mean sea level. This height corresponds to a 10% probability of being exceeded by sea waves in a year. What is the probability of damage to the structure within 3 years?

Problem 4 [15pt, 5pt each]

Hydraulic engineers need to estimate the annual maximum flood flow exceeded within a 0.2% probability (a chance of 1 in 500) for a reservoir's spillway. If such floods have a Gumbel distribution with mean 1500 cms and standard deviation 500 cms.

- (a) What is the design value they seek?
- (b) Why is a Gumbel distribution a reasonable choice for describing floods?
- (c) Calculate the return period of the mean flood under the Gumbel distribution

Problem 5 [10pt]

Refer to a flow data set of a certain region which has the following observations (unit cms) from two sites A and B:

- Site A: 7, 9, 2, 16, 11, 19, 5, 10, 3, 8
- Site B: 11, 26, 33, 19, 16, 20

Assuming that the region is homogeneous in distribution, estimate the 10-year flood for an ungauged site C. What kind of reasonable assumptions do you make for the distribution?

~ Good Luck! ~